

Investigating the Application of a Recommender System for Irish College Courses

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In 2024, over 48,000 Irish secondary school students progressed to college after graduation, whereas in 2018, about 10,000 Irish graduates reported they were not likely to study their college course again, demonstrating the importance and difficulty in choosing a relevant college course in Ireland. This dissertation proposes a Recommender System (RS) that provides users with relevant Irish college course recommendations, assisting students in this difficult decision-making process. The RS is a hybrid RS with Knowledge-Based Filtering and Content-Based Filtering, utilising a student's academic interests, RIASEC model, expected Leaving Certificate (LC) points, NFQ level and college preferences to provide 20 course recommendations. This RS was evaluated in an experiment with $n = 31$ participants, where participants blindly evaluated the relevance of recommendations provided by this RS and a baseline RS, which returns 20 courses sorted by LC points. Large, statistically significant ($p \leq 0.05$) improvements were observed across the precision, serendipity and trust metrics, and insignificant differences for the diversity metric. Moreover, participants liked four out of 20 baseline recommendations on average, all four of which were not on their original college application list. Alternatively, participants liked nine out of 20 actual recommendations, eight of which were not on their original college application list, demonstrating the potential effectiveness of an RS for Irish college courses.